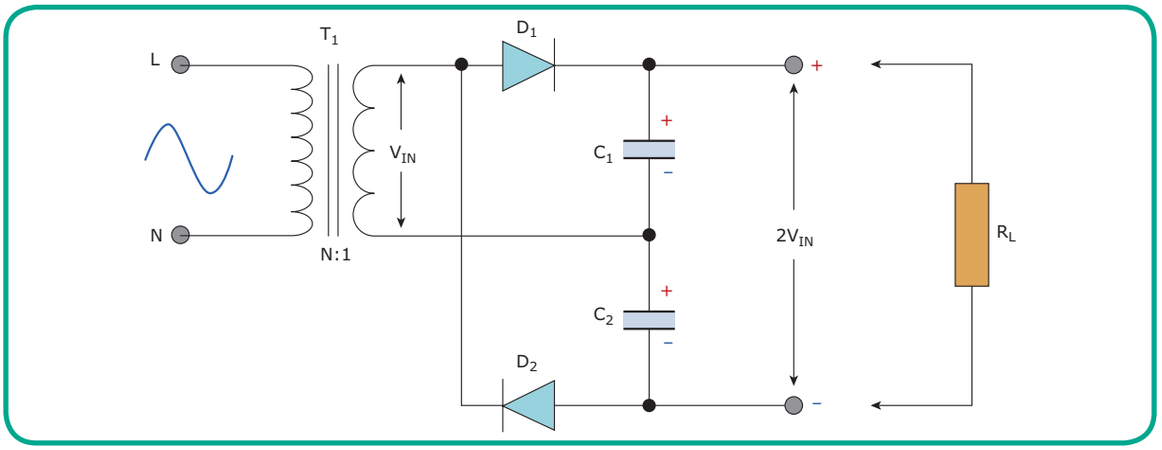


Figure 4.22 Voltage Doubler

When the secondary voltage is positive, the first diode D_1 is forward biased and the primary capacitor (C_1) charges to approximately Vp. During the negative half cycle, the second diode D_2 is forward biased and the secondary capacitor (C_2) charges to approximately Vp. Therefore, the output voltage 2Vp is taken across the two capacitors in series.

Thus the output voltage is double as that of the input AC voltage. It is important to note that the voltage rating of the capacitor should be higher than the AC voltage.

4.10. AC ADAPTER

Figure 4.23 shows an AC adapter. An AC adapter (AC/DC adapter) or (AC/DC converter) is a power supply built into an AC main power plug. AC adapters are also known by various other names such as plug-pack or plug-in-adapter or wall-wart.

DO YOU KNOW?

What type of charger is used in mobile phone?

The method of power supply employed in mobile phone charger is Switched Mode Power Supply (SMPS)

AC adapters typically have a single AC or DC output that is conveyed over a hard-wired cable to a connector, but some adapters have multiple outputs that may be conveyed over one more cables. "Universal" AC adapters have interchangeable input connectors to accommodate different AC mains voltages.

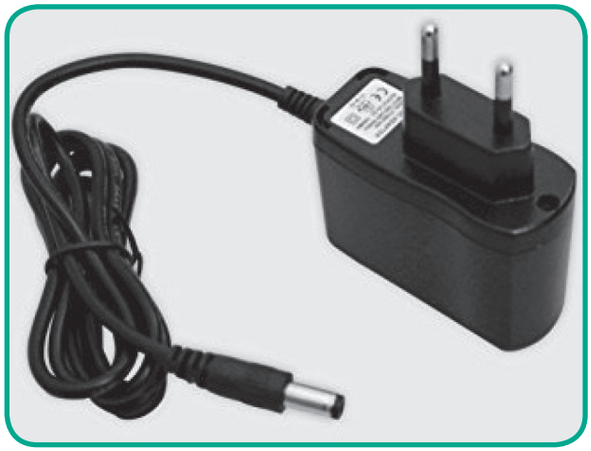


Figure 4.23 AC Adapter

Adapters with AC outputs may consist of a passive transformer (plus a few diodes in DC output adapters) or they may employ switched-mode circuitry. AC adapters consume power and produce electric and magnetic fields even when not connected to a load.

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